

# AIAP Pilot and Validation Protocol

*Preregistration-ready design for testing the load-bearing mechanisms*

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## Evidence boundary

This protocol prepares a governed empirical study. It does not constitute preregistration, ethics approval, institutional authorization, or validation. Local teams must complete estimands, thresholds, sample-size analysis, approvals, governance, and analysis code before data collection.

### 1. Primary research questions

#### RQ1 — Reliability and decision consistency

Can trained assessors score the Lane 2A authentication event consistently enough for the intended use, and what classification errors occur at the decision boundary?

**Primary score statistic:** two-way random-effects, absolute-agreement ICC for a single assessor, with confidence interval and variance components.

**Primary decision evidence:** raw agreement, cross-classification, false-positive and false-negative consequences, and classification consistency at the preregistered cut.

**Sensitivity analyses:** quadratic-weighted kappa and Gwet's AC2, reported with prevalence and category counts. Because Vach and Gerke (2023) analyse AC1 rather than AC2, their critique is treated as a caution about coefficient properties, not direct validation or rejection of AC2. Prompt/form exposure, assessor-session order, fatigue, and drift are prespecified diagnostics. No statistic may be selected after outcomes are known.

#### RQ2 — Incremental and generalization validity

Does authentication add a preregistered practically important increment over product-only evidence, converge with an independent criterion addressing the same construct, and generalize across prompts/cases, occasions, raters, and modalities?

Use a secured unseen transfer task where feasible. Present demonstrable competence is a declared profile of explain, defend, verify, adapt, and transfer; programmes identify the load-bearing components for each claim. Estimate subscore correlations and internal structure, and examine factor structure only where the sample and design support it. Do not treat the profile as one latent trait by default.

A generalizability study or cross-classified mixed model should estimate student, case/prompt, form, occasion, rater, modality, exposure status, and interaction variance. Conduct a decision study when the design supports it. Record the submission-authentication interval and test its association with performance and mismatch.

#### RQ3 — Feasibility, equity, privacy, and decision utility

Can the mechanism be delivered within the Assurance Capacity Gate without disproportionate subgroup burden, construct-irrelevant oral-performance variance, privacy harm, accessibility failure, or negligible decision value?

### 2. Secondary and exploratory questions

Secondary outcomes may include student comprehension, interpretive load, assessment anxiety, disclosure behaviour, agency, evaluative judgement, tool access, marker experience, programme-map completeness, Lane 2B drift, agentic-AI resilience, and comparative utility against simpler frameworks. These outcomes must not be promoted to primary after results are known.

### 3. Design

A minimum Phase 1 design should include one or two programmes, two or three modules, approximately 120 students as a feasibility starting point rather than a power claim, at least two cases/prompts or occasions for a planned subsample, independent double rating, and a random triple-rated subset. Phase 2 should then expand to at least two and preferably three or more institutions with different profiles, retain the preregistered core measures and realistic comparators, and collect evidence over at least two academic cycles to test transportability, comparative utility, and programme-level claims.

Sample size shall be derived from the primary estimand and precision target using validated software or simulation. Subgroup findings that do not meet the precision target remain exploratory; a non-significant difference is not evidence of equivalence.

### 4. Authentication event

Map each prompt to the declared construct and the load-bearing competence components. Include at least one materially unseen or meaningfully perturbed requirement. A five-to-seven-minute structured event may be tested, but lower-burden alternatives—supervised reconstruction, staged sign-off, live walkthrough, or asynchronous evidence with selective live follow-up—should be compared where feasible.

The scoring guide must separate disciplinary understanding from confidence, accent, speed, extroversion, or presentation style unless oral communication is itself part of the construct.

#### 4.1 Prompt exposure and form-equivalence study

Preregister the cohort, exposure window, prompt source, allocation, rotation, retirement, and exposure-monitoring rules. Prefer live artifact-specific perturbations where feasible. Record prompt and form IDs, use counts, suspected circulation, retirement, and affected decisions. Before parallel forms carry the same hurdle, establish a common blueprint and collect expert-review, difficulty, decision-consistency, and accessibility evidence. Treat prompt/form as a facet in the reliability and generalization analyses where the design permits.

### 5. Independent criterion evidence

Collect an external criterion for the same capability, preferably a short secured unseen transfer task scored by assessors blinded to product and authentication results. Declare construct coverage, reliability, timing, contamination risk, and expected relation. The criterion is not a perfect gold standard.

### 6. Standard setting

For any hurdle, preregister a defensible method such as modified Angoff, contrasting groups, or borderline regression where appropriate. Record judge selection, training, uncertainty, review, classification consistency, and consequences at the cut.

### 7. Equity and accessibility

Preregister practically important subgroup differences with stakeholder justification. Estimate effects and uncertainty for EAL status, accommodation status, anxiety, tool access, modality, and alternative-format use. Test whether alternatives address the same construct. Record accessibility incidents, rescheduling, complaints, appeals, and abandonment. Include at least one worked equivalent route, such as a supervised written unseen-transfer task with selective live follow-up, scored against the same construct map. Do not infer equivalence from format similarity.

## 8. Modality and adversarial testing

Where remote authentication is used, conduct a staff-only red-team attempt to defeat the event through plausible covert real-time assistance under ethics and security controls. Also test AI-rehearsed defence. Rehearsal itself is legitimate preparation; the event fails when rehearsed or covertly assisted performance can satisfy the claim without genuine adaptation, verification, or transfer.

## 9. Workload and burden

Record one-off and recurring staff-hours, elapsed scheduling time, assessor throughput, transition time, maximum continuous session length, breaks, moderation, calibration and drift checks, accessibility, rescheduling, recording review, appeals, data governance, and incident handling. Record student preparation, waiting, travel/setup, event, accessibility, rescheduling, and appeal time. Configure a local comparator for the assessment route being replaced or supplemented; compare decision utility, construct coverage, reliability, privacy, equity, and burden, not cost alone.

### 9.1 Group assessment and individual attribution

Where a shared artifact is used, record whether the claim is group-level or individual. Individual certification requires separately attributable evidence and a separate decision for every student. Group-level evidence must not be substituted for missing individual evidence.

### 9.2 Capacity-degradation events

Record assessor illness, mass rescheduling, technology failure, prompt compromise, recording loss, accessibility failure, and data incidents. State the fallback used, affected students and claims, delay or narrowing, and requalification decision.

## 10. Data governance

Complete a data inventory and local legal/ethics review. Minimize recordings and process traces. State access, retention, correction, deletion, incident response, and publication rules. Protect accommodation and disability data. Publish null and adverse results without author, sponsor, candidate, or host veto.

## 11. Preregistered continuation rules

Before data collection, state thresholds or minimum practically important differences that lead to continuation, redesign, narrower claims, replacement, pause, or rejection. A procedurally complete mechanism must still be rejected or narrowed when incremental validity or decision utility is negligible relative to burden.

## 12. Reporting

Report all primary and sensitivity analyses, missingness, deviations, adverse events, subgroup precision, workload, privacy/accessibility outcomes, and the full decision record. Do not describe a feasibility signal as validation.

## Appendix A. Minimum dataset fields

| Domain         | Fields                                                                                                                                                                                                                               |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Participant    | pseudonymous ID, programme/module, declared subgroup variables under approved governance                                                                                                                                             |
| Assessment     | claim ID, lane, modality, prompt/case, occasion, product score                                                                                                                                                                       |
| Authentication | event ID, assessor and session IDs, component scores (explain, defend, verify, adapt, transfer), total/decision, duration, submission-authentication interval, prompt/form ID, prompt source, exposure status, unseen/perturbed item |
| Criterion      | independent criterion score/decision, timing, assessor blinding                                                                                                                                                                      |
| Burden         | staff role hours, transition and breaks, assessor drift, student preparation/wait/travel/event time, scheduling, rescheduling, accessibility, appeal                                                                                 |
| Governance     | consent/notice basis, access, retention, prompt exposure/retirement, capacity incident, appeal, withdrawal/correction                                                                                                                |

## Appendix B. Analysis decision table

| Question | Primary estimand/evidence                                                            | Binding decision use                                                         |
|----------|--------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| RQ1      | ICC absolute agreement; raw decision agreement and error consequences                | whether scoring/decisions are sufficiently stable                            |
| RQ2      | incremental validity and generalization variance                                     | whether the claimed inference is supported and how much sampling is required |
| RQ3      | capacity gate, subgroup estimates, privacy/accessibility incidents, decision utility | whether the mechanism is feasible and proportionate                          |